GRE Quantitative Reasoning

모의고사 Set 9 — 문제지

최신 문제(뒷번호) · S2 고난도

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내
- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 9 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
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Section 1 12 Questions · 21 Minutes

Question 1 대소비교

1136. Quantity A: The number of integer solutions of the equation $3x^2+5x-2=0$
Quantity B: 1

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

501. When $2(10^{100})+1$ is divided by 3, the remainder is r.
Quantity A: r
Quantity B: 1

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

1247. x is the units digit of the number $11^{14}-1$.
Quantity A: x
Quantity B: 1

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

950. Let x be a positive number. The random variable R is normally distributed with mean x and standard deviation 0.25x. The random variable T is normally distributed with mean 0.5x and standard deviation 0.5x.
Quantity A: P (R <1.375x)
Quantity B: P (T <1.25x)

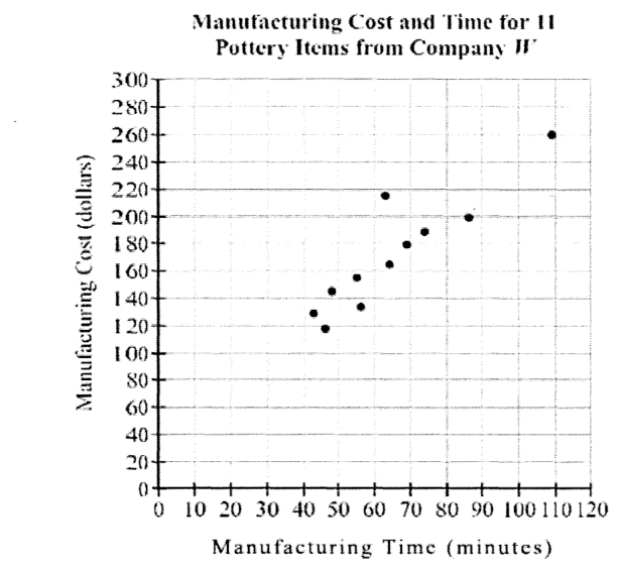
(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 객관식

403. A certain box is in the shape of a cube. If the total surface area of the box is 384 square feet, what is the total length, in feet, of all the edges of the box?
A48
B64
C72
D80
E96

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Questions 6 to 8 are based on the following data.



Question 6 객관식

994. If one item is to be randomly selected from the items whose manufacturing cost is greater than $140, what is the probability that the item selected will be one whose manufacturing time is greater than 60 minutes?
A3/8
B1/2
C5/8
D2/3
E3/4

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 7 객관식

995. For each item, a manager calculates the ratio of the manufacturing cost to the manufacturing time. Which of the following is closest to the value of the greatest of these eleven ratios, in dollars per minute?
A2.5
B3.5
C4.0
D4.5
E5.0

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 8 객관식

996. The manufacturing cost of the item that takes the most time to manufacture is approximately what percent greater than the cost of the item that takes the least time to manufacture?
A10%
B50%
C75%
D100%
E200%

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 9 객관식

892. Machines A, B, and C produce widgets at their respective constant rates. Machine A produces n widgets in x hours, machine B produces n widgets in 2x hours, and machine C produces n widgets in 3x hours. In terms of x, how many hours does it take the three machines, working simultaneously at their respective constant rates to produce n widgets?
A(1/11)x
B(1/6)x
C(6/11)x
D(11/6)x
E2x

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 10 주관식

927. The 25 students in an English class took a literature test. Jovan, one of the 25 students, originally received a score on the test that was 6.2 points above the average (arithmetic mean) score for the 25 students. Each student was then given an opportunity to earn 10 additional points on the test by completing an extra project, and the average was then recalculated. If Jovan earned the 10 additional points and no other student earned additional points, Jovan's new score was how many points above the recalculated average score?

Question 11 복수선택

1183. The weights, in ounces, of the two candles in package A are 4.00 and 4.20. The weights, in ounces, of the four candles in package B are 3.80, 4.00. 4.20, and 4.40. Two candles are to be added to package A such that the standard deviation of the weights of the four candles in package A will be less than the standard deviation of the weights of the four candles in package B. Which of the following pairs of values could be the weights, in ounces, of the two candles to be added to package A?
A.3.70 and 4.50
B.3.90 and 4.30
C.4.00 and 4.20
D.4.05 and 4.15

Indicate all that apply. A B C D

Question 12 객관식

1152. If 12 percent of the books in a library are reference books, then the ratio of the number of reference books to the number of nonreference books is closest to which of the following?

A.1 to 6
B.1 to 7
C.1 to 8
D.1 to 10
E.1 to 12

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Section 2 15 Questions · 26 Minutes

Question 1 대소비교

1131. In plane P, point Q is on line l.
Quantity A: The number of points in plane P whose distance from line l is 2 and whose distance from point Q is 3
Quantity B: 4

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

579. x is an integer, and $1/9 < 3^x < 90$.
Quantity A: The number of possible values of x
Quantity B: 6

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

695. There are 7 balls in a box, among which 1 is red. Someone removes balls from the box one by one without replacement and stops until he or she removes the red ball.
Quantity A: The probability that he or she stops after the third removal
Quantity B: The probability that he or she stops after the fourth removal

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

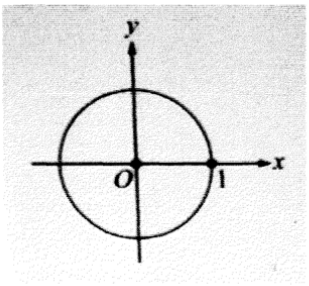
Question 4 대소비교

554. The probability that event R will occur is 0.65. The probability that event T will occur is 0.47.
Quantity A: The probability that neither R nor T will occur
Quantity B: 0.47

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 대소비교

594.



In the rectangular coordinate plane, (x, y) is a point on the circle with center O and radius 1, and $xy \neq 0$.
Quantity A: $|x^3+y^3|$
Quantity B: 1

(A) Quantity A is greater.

(B) Quantity B is greater.

(C) The two quantities are equal.

(D) The relationship cannot be determined from the information given.

Question 6 객관식

1007. Cylindrical can A contains five servings of soup when full. How many servings of the soup are contained in a full cylindrical can with 2 times the diameter and 3 times the height of can A?
A30
B60
C90
D120
E210

(A) (B) (C) (D) (E)

Question 7 객관식

880. What is the units digit of the sum $13^{10}+17^{10}$?
A0
B2
C4
D6
E8

(A) (B) (C) (D) (E)

Question 8 객관식

1261. Three 16-ounce bottles, M, R,and S, are filled with three different beverages. The amount of sugar contained in bottle M is 1/4 of the amount of sugar contained in bottle R and is 2/5 of the amount of sugar contained in bottle S. The amount of sugar contained in bottles R and S combined is what fraction of the amount of sugar contained in bottle R?
A.5/8
B.7/8
C.9/8
D.11/8
E.13/8

(A) (B) (C) (D) (E)

Question 9 복수선택

703. The integer n is the product of five consecutive positive integers. Which of the following integers must be a factor of n?
Indicate all such numbers.
A2
B3
C4
D5
E8
F9
G10

Indicate all that apply. A B C D E F G

Question 10 주관식

1245.

Textbook	Number of Pages
A	510
B	480
C	490
D	520
E	x

The table shows the number of pages in each of 5 textbooks. What is the greatest possible value of x for which the average (arithmetic mean) number of pages of the 5 textbooks is equal to the median number of pages of the 5 textbooks?

Question 11 객관식

469. A certain desk calendar shows the number of the day of the year and the number of days remaining in the year. If the calendar shows 1/364 for Saturday, January 1, what does the calendar show for the Tuesday that is 20 weeks after the first Tuesday in January?
A21/344
B141/224
C142/223
D143/222
E144/221

A B C D E

Question 12 객관식

771. A box contains 100 purple balls, 100 red balls, and 100 white balls. What is the minimum number of balls that must be chosen to ensure that at least 4 of the balls chosen have the same color?
A5
B9
C10
D12
E13

A B C D E

Question 13 복수선택

1259.

Number of Tickets	Cost
1	$1.25
10	$11.00
20	$20.00

Tickets for a carnival are sold either individually or in packages of 10 tickets or 20 tickets at the costs shown in the table. A group of friends bought n tickets for the least possible total cost. A second group of friends bought more than n tickets for a total cost that was less than the first group`s total cost. Which of the following could be the value of n?
Indicate all such values.

A.8
B.9
C.11
D.12
E.16
F.17
G.18
H.19

Indicate all that apply. A B C D E F G H

Question 14 객관식

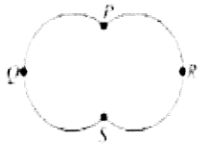
825. In a local election there are 2 candidates for mayor, 4 candidates for sheriff, and 5 candidates for dogcatcher on the ballot. In each of the three categories a voter may vote for exactly one candidate or none. How many different ways can a voter fill out the ballot?

A11
B14
C40
D80
E90

A B C D E

Question 15 객관식

1310.



The figure represents the shape of a certain swimming pool. If PQS and PRS are arcs of two circles, each of radius 3 meters, and the center of each circle is on the circumference of the other, what is the perimeter in meters, of the swimming pool?

A.4 π
B.6 π
C.8 π
D.12 π
E.18 π

A B C D E
